



Household Energy Poverty and Individual Time Allocation—Empirical Evidence from China

Shang Xie Peking University

Zhiquan Li Southwestern University of Finance and Economics

Hui Zhang Peking University

Dongming Wei Peking University

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Household energy poverty refers to household that lack access or ability to use sufficient, secure, clean, and environmentally friendly modern energy (Charlier, 2019; Dogan et al., 2022; Luan et al., 2023).

Individual time allocation refers to people how to allocate their daily time work, housework, and leisure.

Household energy poverty may affect the individual time allocation

Literature Review- The measure of household energy poverty

Accessibility

The household is classified as suffering from energy poverty if the main cooking fuel is firewood or coal (Adusah-Poku and Takeuchi, 2019; Zhang et al., 2019; Luan et al., 2023).

Affordability

The household is classified as suffering from energy poverty if ratio of the expenditure to the household income is greater than 10% (Boardman, 1991; Barnes et al., 2011; Bouzarovski and Petrova, 2015; Churchill and Smyth, 2020) .

Multidimensional

Combine affordability, accessibility, appliances, building efficiency and other factors to measure energy poverty (Zhang et al., 2019; Abbas et al., 2022; Luan et al., 2023).

Literature Review- The influence of energy poverty

Health

Energy poverty can significantly decrease health (Zhang et al., 2021; Churchill & Smyth, 2021).

Income inequality

Energy poverty can significantly increase income inequality (Igawa, & Managi, 2022).

Education

Children may face difficulties while studying in the evening due to lack of electricity, which may further affect their academic performance (Apergis et al., 2022).

Entrepreneurship

Accessibility energy poverty and affordability energy poverty increase the likelihood of entrepreneurship (Cheng et al., 2021).

Subjective well-being

Energy poverty can significantly decrease individual's subjective well-being (Igawa & Managi, 2022).

Literature Review- The influential factors of individual time allocation

Macro perspective

The overall level of economic development of society, the social employment situation, government employment policy control, the impact of health emergencies, and traditional cultural can affect individual time allocation(Hersch and Stratton, 2002; Zhang Jinhua and Hu Junhui, 2012; Chen and Li, 2014).

Micro perspective

Individual characteristics are gender, age, education level, non-cognitive ability, and property of work (Becker, 1960; 1976; Hersch and Stratton, 2002)

Household characteristics familysize, number of children, gender of the children, whether it supports the elderly, whether there is intergenerational economic help, intergenerational rearing and caring (Mankiw, 2014 ; Guo et al, 2018; Sun, 2021; Guo, Kaiming et al., 2021).



- Regarding family factors that affect personal time allocation, although energy poverty at the family level is also one of the important characteristics of families, there are few studies on the impact of family energy poverty on individual time allocation.
- Individual time allocation highly correlated to the population development and economics development, thus we conduct this study to find the relationship between household energy poverty and individual time allocation.

Data Source: China Family Panel Studies (CFPS) 2014, 2016, 2018

Table 1 Summary Statistics

Variables	Description	Mean	Std. Dev	Obs
Independent variables				
Working Time	Working hours per week	30.19	27.90	48619
Housework Time	Housework time per week	9.00	12.00	57112
Leisure Time	Leisure time per week	12.63	15.34	57112
Dependent variable				
FMEPI	Multidimensional measure of household energy poverty.			
	FMEPI =1, no firewood or coal for cooking, ratio of energy consumption to total income less than 10%, household income greater than median income	1.54	0.64	57112
	FMEPI =3, cooking with firewood or coal, ratio of energy consumption to total income greater than 10%, household income less than median income			
	FMEPI =2, otherwise			

Individual characteristics				
Gender	Male=1, Female=0	0.51	0.50	57112
Age	Age (in years)	39.40	13.60	57112
Hukou	Type of registered permanent residence. Non-rural=1, rural=0.	0.27	0.44	57556
Education	Years of education	7.74	4.69	57556
Health	Health status observed by interviewers (very bad 1-2-3-4-5-6-7 very good)	5.71	1.14	36635
Household characteristics				
Family Size	Number of family members	4.49	1.94	57112
F_income	Average annual household income per capita (in RMB 10,000)	1.44	1.44	57112
House	Whether the family has limited property rights or full property rights of a house. Fully=1, otherwise=0	0.88	0.32	57112
Asset	Household net assets (in RMB 10,00)	55.44	125.25	55186
Proportion16	Proportion of children aged 16 and below in the household	2.55	1.44	57112
Proportion65	Proportion of the elderly aged 65 and above in the household	0.09	0.18	57112

Note: The data presented above consists of longitudinal cross-sectional data from the China Family Panel Studies (CFPS) for the years 2014, 2016, and 2018.

Descriptive Figure

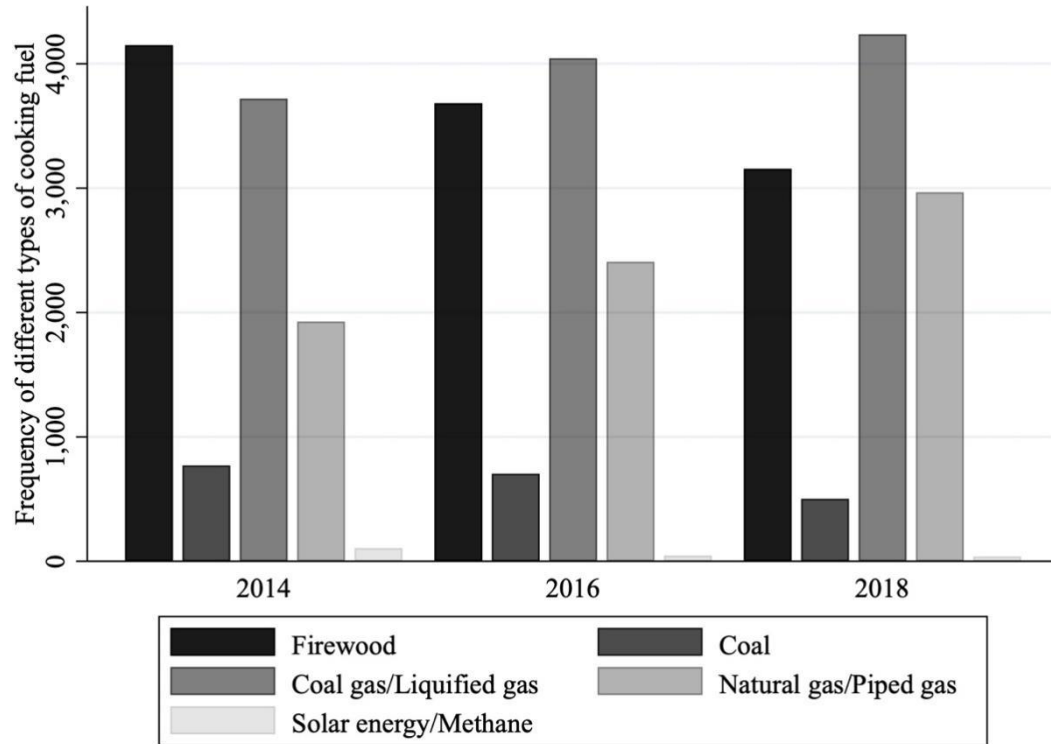


Fig. 1. The frequency of different types of fuel used for cooking in China (2014-2018)

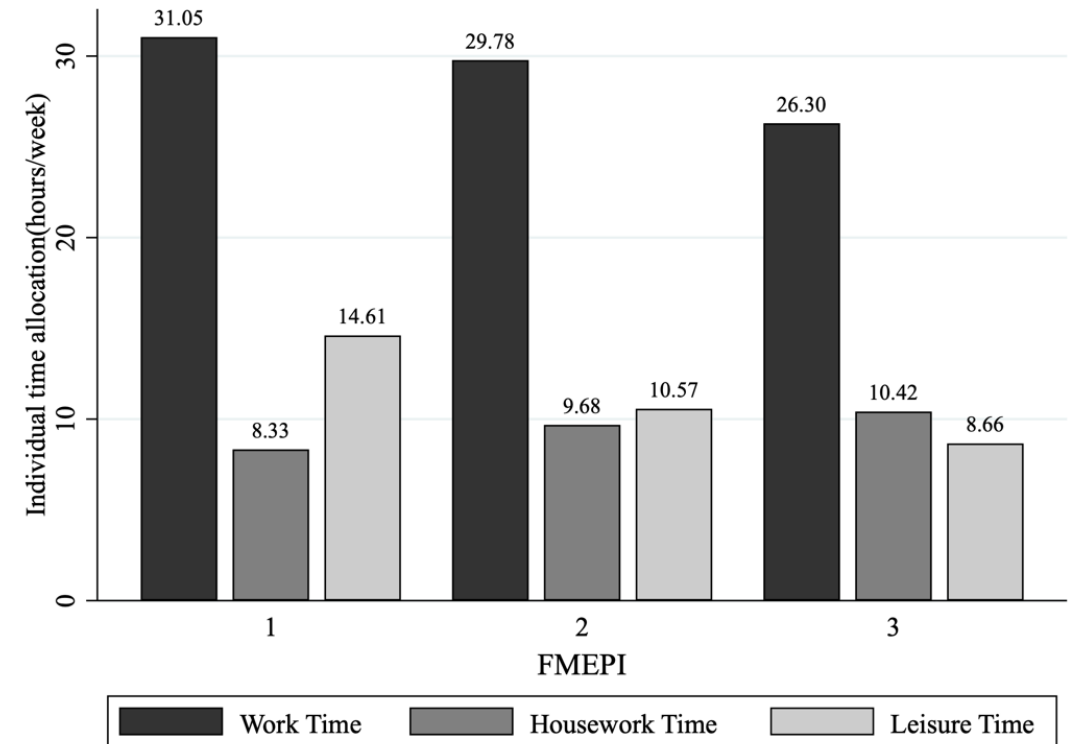


Fig. 2. The relationship between FMEPI and individual time allocation (2014-2018)

Identification equation

$$Time_{cit} = \alpha + \beta FMEPI_{cit} + \gamma \mathbf{X}_{cit} + \mu_{ih} + \tau_t + \varepsilon_{cit}$$

where the subscript $i = 1, 2, \dots, N$ represents individual; $c = 1, 2, \dots, N$ refers to the community the individual live; $t = 2014, 2016, 2018$ indicates the time of data collection.

$Time_{cit}$ stands the individual i 's time allocation who lived in community c in year t , including weekly working time, weekly housework time, and weekly leisure time.

$FMEPI_{cit}$ denotes the family multidimensional household energy poverty index of individual i who lived in community c in year t .

$\mathbf{X}_{cit}$ is a vector of control variables including individual characteristics (Gender, Age, Hukou, Education, Health) and household characteristics (Family Size, F_income, House, Asset, Proportion16, Proportion65).

μ_{ih} signifies the county fixed effect, τ_t refers to the year fixed effect, ε_{cit} denotes the idiosyncratic error term.

β captures the response of individual working time, housework time, and leisure time to household energy poverty.

Baseline regression result

Table 2

Energy Poverty and Individual Working Time, Housework Time, and Leisure Time (Baseline OLS regression).

Variables	Working Time		Housework Time		Leisure Time	
	(1)	(2)	(3)	(4)	(5)	(6)
FMEPI	-0.779*** (0.242)	-0.729*** (0.248)	1.000*** (0.119)	0.962*** (0.122)	-0.871*** (0.142)	-0.780*** (0.145)
Individual	Yes	Yes	Yes	Yes	Yes	Yes
Household	No	Yes	No	Yes	No	Yes
Year FE	Yes	Yes	Yes	Yes	Yes	Yes
County FE	Yes	Yes	Yes	Yes	Yes	Yes
R-squared	0.077	0.079	0.223	0.222	0.125	0.128
Observations	29276	28585	36635	35802	36635	35802

Note: All columns in the table present regression results with robust standard errors at the county level. Year FE refers to year fixed effect. County FE refers to county fixed effect. * $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$.

- OLS regression result shows that FMEPI has significantly negative impact on working time, significantly positive impact on housework time, significantly negative impact on leisure time.
- However, it may face endogeneity problems: (1) unobservable omitted variables (2) reverse causality problem.

Table 3

Regression results with CEP and AEPS as instrumental variables (2SLS regression).

Variables	Working Time		Housework Time		Leisure Time	
	(1)	(2)	(3)	(4)	(5)	(6)
FMEPI		-2.995*** (0.727)		3.782*** (0.349)		-3.071*** (0.450)
CEP	-0.041** (0.017)		-0.035** (0.016)		-0.035** (0.016)	
AEPS	0.672*** (0.012)		0.690*** (0.011)		0.690*** (0.011)	
Individual	Yes	Yes	Yes	Yes	Yes	Yes
Household	Yes	Yes	Yes	Yes	Yes	Yes
Year FE	Yes	Yes	Yes	Yes	Yes	Yes
County FE	Yes	Yes	Yes	Yes	Yes	Yes
Hansen J P-value		0.443		0.628		0.767
Wald chi2 statistics	1742.39		2221.14		2221.14	
F statistics of first stage	1483.90		1903.35		1903.35	
R ²		0.078		0.209		0.123
Observations	27750	27750	34848	34848	34848	34848

Note: Wald chi2 statistics refers to Cragg-Donald Wald F statistic. F statistics of first stage refers to Kleibergen-Paap Wald F statistic. The Individual includes gender, age, education, hukou, health. The Family includes family size, house, F_income, asset, proportion16, proportion65. All columns in the table present regression results with robust standard errors at the county level. Year FE refers to year fixed effect. County FE refers to county fixed effect. * $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$.

- Use the Clean Energy Provincial Demonstration Policy (CEP) as instrumental variable for accessibility (Ningxia in 2012; Zhejiang in 2014; Sichuan in 2016; Tibet in 2016; Gansu in 2016; and Qinghai in 2018). CEP=1; otherwise, CEP=0
- Use the average energy poverty share of the community (AEPS, excluding the focal household) is used as instrumental variable for affordability

Table 4

Cooking Fuel Types and Individual Working Time, Housework Time, and Leisure Time (OLS regression).

Variables	Working Time	Housework Time	Leisure Time
	(1)	(2)	(3)
EP	-0.493 (0.339)	1.333*** (0.164)	-1.286*** (0.199)
Individual	Yes	Yes	Yes
Family	Yes	Yes	Yes
Year FE	Yes	Yes	Yes
County FE	Yes	Yes	Yes
R ²	0.079	0.223	0.129
Observations	28514	35719	35719

Note: The Individual includes gender, age, education, hukou, health. The Family includes family size, house, F_income, asset, proportion16, proportion65. All columns in the table present regression results with robust standard errors at the county level. Year FE refers to year fixed effect. County FE refers to county fixed effect. * $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$.

Alternative measure of energy accessibility poverty:

A household is defined as energy poverty if it uses firewood or coal for cooking, EP = 1, otherwise 0.

Table 5

Energy Poverty Level and Individual Working Time, Housework Time, and Leisure Time (OLS regression).

Variables	Working Time		Housework Time		Leisure Time	
	15%	20%	15%	20%	15%	20%
	(1)	(2)	(3)	(4)	(5)	(6)
Energy Poverty	-0.796***	-0.808***	0.914***	0.917***	-0.729***	-0.763***
Line	(0.267)	(0.281)	(0.132)	(0.138)	(0.157)	(0.165)
Individual	Yes	Yes	Yes	Yes	Yes	Yes
Family	Yes	Yes	Yes	Yes	Yes	Yes
Year FE	Yes	Yes	Yes	Yes	Yes	Yes
County FE	Yes	Yes	Yes	Yes	Yes	Yes
R ²	0.079	0.079	0.222	0.222	0.128	0.128
Observations	28585	28585	35802	35802	35802	35802

Note: The Energy Poverty Level includes 15% and 20%. The Individual includes gender, age, education, hukou, health. The Family includes family size, house, F_income, asset, proportion16, proportion65. All columns in the table present regression results with robust standard errors at the county level. Year FE refers to year fixed effect. County FE refers to county fixed effect. * $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$.

Alternative threshold of energy affordability poverty :

A household is defined as energy poverty if the proportion spend more than 15% or even 20% of their household income on energy.

Table 6

Energy Poverty and Individual Working Time, Housework Time, and Leisure Time (Tobit regression).

Variables	Working Time	Housework Time	Leisure Time
	(1)	(2)	(3)
FMEPI	-0.729*** (0.235)	1.111*** (0.134)	-0.876*** (0.153)
Individual	Yes	Yes	Yes
Family	Yes	Yes	Yes
Year FE	Yes	Yes	Yes
County FE	Yes	Yes	Yes
χ^2	2641.38	10194.33	5677.94
Observations	28585	35802	34802

Note: The Individual includes gender, age, education, hukou, health. The Family includes family size, house, F_income, asset, proportion16, proportion65. All columns in the table present regression results with robust standard errors at the county level. Year FE refers to year fixed effect. County FE refers to county fixed effect. * $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$.

Alternative regression model :

Because the dependent variables is truncate data left.

Heterogenous Analysis

Table 7

Gender heterogeneity analysis.

Variables	Working Time		Housework Time		Leisure Time	
	Female	Male	Female	Male	Female	Male
	(1)	(2)	(3)	(4)	(5)	(6)
FMEPI	-0.707*	-0.808**	0.469**	1.489***	-0.847***	-0.734***
	(0.381)	(0.327)	(0.182)	(0.161)	(0.204)	(0.205)
Individual	Yes	Yes	Yes	Yes	Yes	Yes
Family	Yes	Yes	Yes	Yes	Yes	Yes
Year FE	Yes	Yes	Yes	Yes	Yes	Yes
County FE	Yes	Yes	Yes	Yes	Yes	Yes
R ²	0.084	0.060	0.164	0.128	0.149	0.109
Observations	13051	15534	17528	18274	17528	18274

Note: The Individual includes gender, age, education, hukou, health. The Family includes family size, house, F_income, asset, proportion16, proportion65. All columns in the table present regression results with robust standard errors at the county level. Year FE refers to year fixed effect. County FE refers to county fixed effect. * $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$.

FMEPI increases more housework time of man compared with women.

Heterogenous Analysis

Table 8

Age heterogeneity analysis.

Variables	Working Time		Housework Time		Leisure Time	
	<45	>=45	<45	>=45	<45	>=45
	(1)	(2)	(3)	(4)	(5)	(6)
EP	-0.966***	-0.072	0.964***	1.171***	-1.162***	-0.913***
	(0.331)	(0.376)	(0.155)	(0.192)	(0.211)	(0.193)
Individual	Yes	Yes	Yes	Yes	Yes	Yes
Family	Yes	Yes	Yes	Yes	Yes	Yes
Year FE	Yes	Yes	Yes	Yes	Yes	Yes
County FE	Yes	Yes	Yes	Yes	Yes	Yes
R ²	0.071	0.092	0.222	0.195	0.148	0.129
Observations	15829	12756	19367	16435	19367	16435

Note: The Individual includes gender, age, education, hukou, health. The Family includes family size, house, F_income, asset, proportion16, proportion65. All columns in the table present regression results with robust standard errors at the county level. Year FE refers to year fixed effect. County FE refers to county fixed effect. * $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$.

FMEPI reducing working time mainly occurs among the youth group. The youth group as the dominant group of the social development, but FMEPI reduces their working time or job opportunities, which is an issue that deserves attention

Heterogenous Analysis

Table 9

Urban-rural heterogeneity analysis.

Variables	Working Time		Housework Time		Leisure Time	
	Rural	Urban	Rural	Urban	Rural	Urban
	(1)	(2)	(3)	(4)	(5)	(6)
FMEPI	0.029 (0.328)	-0.536\$ (0.417)	0.706*** (0.168)	0.772*** (0.187)	-0.721*** (0.188)	-0.553** (0.251)
Individual	Yes	Yes	Yes	Yes	Yes	Yes
Family	Yes	Yes	Yes	Yes	Yes	Yes
Year FE	Yes	Yes	Yes	Yes	Yes	Yes
County FE	Yes	Yes	Yes	Yes	Yes	Yes
R ²	0.111	0.067	0.208	0.227	0.138	0.103
Observations	14925	13352	18038	17351	18038	17351

Note: The Individual includes gender, age, education, hukou, health. The Family includes family size, house, F_income, asset, proportion16, proportion65. All columns in the table present regression results with robust standard errors at the county level. Year FE refers to year fixed effect. County FE refers to county fixed effect. \$=0.198, * $p<0.1$, ** $p<0.05$, *** $p<0.01$.

FMEPI and working time with a coefficient of -0.536 that is statistically different from zero at 19.8% significance level, suggesting that the loss of working time due to FMEPI mainly comes from urban areas



8.1

Mechanism Analysis

$$IM = \alpha + \beta FMEPI_{cit} + \gamma \mathbf{X}_{cit} + \mu_i + \tau_t + \varepsilon_{cit}$$

where IM is the mechanism variable, including attitudes towards life at the subjective level (Life Satisfaction and Future Confidence), as well as personal health, employment and marital status at the objective level

Mechanism Analysis

A healthy psychological condition is a necessary requirement to guide individuals to work and enjoy life in a positive way, so negative attitudes towards life resulting from household energy poverty are likely to be an important channel influencing the time allocation.

Table 10

Mechanism analysis (Life Satisfaction and Future Confidence).

Variables	Life Satisfaction			Future Confidence		
	(1)	(2)	(3)	(4)	(5)	(6)
Panel A: Gender grouping regression						
	All	Female	Male	All	Female	Male
FMEPI	-0.052*** (0.011)	-0.033** (0.015)	-0.071*** (0.015)	-0.041*** (0.010)	-0.035** (0.015)	-0.045*** (0.014)
Individual	Yes	Yes	Yes	Yes	Yes	Yes
Family	Yes	Yes	Yes	Yes	Yes	Yes
Year FE	Yes	Yes	Yes	Yes	Yes	Yes
County FE	Yes	Yes	Yes	Yes	Yes	Yes
R ²	0.065	0.057	0.074	0.064	0.059	0.069
Observations	35802	17528	18274	35802	17528	18274
Panel B: Age grouping regression						
	All	Age<45	45<=Age	All	Age<45	45<=Age
FMEPI	-0.052*** (0.011)	-0.049*** (0.014)	-0.067*** (0.016)	-0.041*** (0.010)	-0.017 (0.013)	-0.057*** (0.017)
Individual	Yes	Yes	Yes	Yes	Yes	Yes
Family	Yes	Yes	Yes	Yes	Yes	Yes
Year FE	Yes	Yes	Yes	Yes	Yes	Yes
County FE	Yes	Yes	Yes	Yes	Yes	Yes
R ²	0.065	0.066	0.077	0.064	0.049	0.066
Observations	35802	19367	16435	35802	19367	16435
Panel C: Urban-rural grouping regression						
	All	Rural	Urban	All	Rural	Urban
FMEPI	-0.052*** (0.011)	-0.044*** (0.014)	-0.067*** (0.018)	-0.041*** (0.010)	-0.028** (0.014)	-0.063*** (0.017)
Individual	Yes	Yes	Yes	Yes	Yes	Yes
Family	Yes	Yes	Yes	Yes	Yes	Yes
Year FE	Yes	Yes	Yes	Yes	Yes	Yes
County FE	Yes	Yes	Yes	Yes	Yes	Yes
R ²	0.065	0.071	0.066	0.064	0.075	0.057
Observations	35802	18038	17351	35802	18038	17351

A healthy body is a prerequisite for normal work and a necessary guarantee for active enjoyment of life. However, household energy poverty not only affects mental health, but also has a significant impact on an individual’s physical health (Zhang et al., 2021)

Table 11

Mechanism analysis (Health Status)

Variables	Health Status	Health Status	Health Status
	(1)	(2)	(3)
Panel A: Gender grouping regression			
	All	Female	Male
FMEPI	-0.071*** (0.011)	-0.082*** (0.016)	-0.063*** (0.015)
Individual	Yes	Yes	Yes
Family	Yes	Yes	Yes
Year FE	Yes	Yes	Yes
County FE	Yes	Yes	Yes
R ²	0.228	0.255	0.200
Observations	35802	17528	18274
Panel B: Age grouping regression			
	All	Age<45	45<=Age
FMEPI	-0.071*** (0.011)	-0.046*** (0.014)	-0.083*** (0.017)
Individual	Yes	Yes	Yes
Family	Yes	Yes	Yes
Year FE	Yes	Yes	Yes
County FE	Yes	Yes	Yes
R ²	0.228	0.199	0.202
Observations	35802	19367	16435
Panel C: Urban-rural grouping regression			
	All	Rural	Urban
FMEPI	-0.071*** (0.011)	-0.048*** (0.015)	-0.063*** (0.017)
Individual	Yes	Yes	Yes
Family	Yes	Yes	Yes
Year FE	Yes	Yes	Yes
County FE	Yes	Yes	Yes
R ²	0.228	0.256	0.217
Observations	35802	18038	17351

Background

Literature Review

Motivation

Data and model

Baseline

Robustness

Heterogenous

Mechanism

Energy poverty especially the limitation access to electricity enable children cannot study easily after dark (Luan et al., 2023), which further impacts their education. Therefore, the mental and physical health and education that effected by household energy poverty may influence the employment, further affect the time allocation.

Table 12
Mechanism analysis (Employment)

Variables	Employment	Employment	Employment
	(1)	(2)	(3)
Panel A: Gender grouping regression			
	All	Female	Male
FMEPI	-0.182*** (0.035)	-0.149*** (0.056)	-0.209*** (0.046)
Individual	Yes	Yes	Yes
Family	Yes	Yes	Yes
Year FE	Yes	Yes	Yes
County FE	Yes	Yes	Yes
R ²	0.289	0.328	0.268
Observations	35802	17358	18274
Panel B: Age grouping regression			
	All	Age<45	45<=Age
FMEPI	-0.182*** (0.035)	-0.015 (0.043)	-0.420*** (0.069)
Individual	Yes	Yes	Yes
Family	Yes	Yes	Yes
Year FE	Yes	Yes	Yes
County FE	Yes	Yes	Yes
R ²	0.289	0.280	0.332
Observations	35802	19367	16309
Panel C: Urban-rural grouping regression			
	All	Rural	Urban
FMEPI	-0.182*** (0.035)	-0.106* (0.055)	-0.132** (0.051)
Individual	Yes	Yes	Yes
Family	Yes	Yes	Yes
Year FE	Yes	Yes	Yes
County FE	Yes	Yes	Yes
R ²	0.289	0.246	0.255
Observations	35802	18030	17269



Mechanism Analysis

In the case of a single individual, they may need to allocate more of their time to housework. However, when an individual is married, the presence of a spouse can facilitate the division of household tasks, thereby introducing household size effects and reducing time inefficiencies.

Table 13			
Mechanism analysis (Single Status)			
Variables	Single Status (1)	Single Status (2)	Single Status (3)
Panel A: Gender grouping regression			
	All	Female	Male
FMEPI	0.189*** (0.030)	0.151*** (0.043)	0.291*** (0.044)
Individual	Yes	Yes	Yes
Family	Yes	Yes	Yes
Year FE	Yes	Yes	Yes
County FE	Yes	Yes	Yes
R ²	0.301	0.224	0.393
Observations	35802	17528	18274
Panel B: Age grouping regression			
	All	Age<45	45<=Age
FMEPI	0.189*** (0.030)	0.219*** (0.045)	-0.063 (0.059)
Individual	Yes	Yes	Yes
Family	Yes	Yes	Yes
Year FE	Yes	Yes	Yes
County FE	Yes	Yes	Yes
R ²	0.301	0.443	0.190
Observations	35802	19367	16422
Panel C: Urban-rural grouping regression			
	All	Rural	Urban
FMEPI	0.189*** (0.030)	0.188*** (0.041)	0.143*** (0.051)
Individual	Yes	Yes	Yes
Family	Yes	Yes	Yes
Year FE	Yes	Yes	Yes
County FE	Yes	Yes	Yes
R ²	0.301	0.320	0.300
Observations	35802	18026	17312
Baseline Robustness Heterogenous Mechanism			



Conclusion and Suggestion

Conclusion

- We found that household energy poverty significantly reduces individual working time and leisure time, but increases the time for housework. Life attitudes, health status, employment, and single status could be potential mechanisms to explain how household energy poverty affect individual time allocation. Our results demonstrate that household energy poverty reduces the effective time allocation of individual, resulting in a waste of human resources, which may hinder social sustainable development.

Suggestion

- Firstly, it is necessary to improve the accessibility to modern energy. Although the coverage of electricity has reached 100% in China, the insufficient and unstable electricity supply problems exist, especially in remote areas.
- Secondly, governments are supposed to pay more attention to the energy poverty among male group according to the heterogeneity analysis for gender.
- Thirdly, the issue of energy poverty among the younger group also demands urgent attention.

THANKS



Alice

北京 海淀



扫一扫上面的二维码图案，加我为朋友。

222120202034@smail.swufe.edu.cn